

Step 1: Importing Python libraries

In [1]:

```
# Importing libraries

import pandas as pd
import numpy as np
import os
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Importing Customed Stock Dataset

In [2]:

```
path = r'/Users/matthewabrams/Desktop/Stocks Data/'
stocks = pd.read_csv(os.path.join(path, 'Stock Datasets/New Files/6_Stocks_Trans
stocks.drop(columns = 'Unnamed: 0', inplace = True)
stocks
```

Out[2]:

	Date	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months
0	2015-01-02	Amazon	55664000	15.4260	15.6290	15.7375	15.3480	NaN
1	2015-01-05	Amazon	55484000	15.1095	15.3505	15.4190	15.0425	NaN
2	2015-01-06	Amazon	70380000	14.7645	15.1120	15.1500	14.6190	NaN
3	2015-01-07	Amazon	52806000	14.9210	14.8750	15.0640	14.7665	NaN
4	2015-01-08	Amazon	61768000	15.0230	15.0160	15.1570	14.8055	NaN
...	...	...	...	...	...	...	...	...
15091	2024-12-24	Tesla	59551800	462.2800	435.9000	462.7800	435.1400	239.391967
15092	2024-12-26	Tesla	76366400	454.1300	465.1600	465.3300	451.0200	240.972459
15093	2024-12-27	Tesla	82666800	431.6600	449.5200	450.0000	426.5000	242.386011
15094	2024-12-30	Tesla	64941000	417.4100	419.4000	427.0000	415.7500	243.700383
15095	2024-12-31	Tesla	76825100	403.8400	423.7900	427.9300	402.5400	244.968579

15096 rows × 9 columns

Step 3: Descriptive Stats

In [3]:

```
stocks.describe()
```

Out [3]:

	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months
count	1.509600e+04	15096.000000	15096.000000	15096.000000	15096.000000	14004.000000
mean	6.103421e+07	176.535332	176.487295	178.860965	174.063377	172.653731
std	6.256008e+07	146.962416	146.941842	148.870957	144.954963	133.289699
min	1.144000e+06	9.578000	9.488000	10.331300	9.403300	14.051845
25%	1.811868e+07	58.673125	58.559225	59.095975	58.012625	63.993417
50%	4.190585e+07	145.340000	145.165000	147.130000	143.215000	150.155574
75%	8.501708e+07	242.650000	242.587500	246.242500	238.835000	242.531544
max	9.140820e+08	936.560000	933.350000	941.750000	923.080000	709.552623

In [4]:

stocks.groupby('Stock_Name').describe().transpose()

Out [4]:

	Stock_Name	Amazon	Apple	Meta	Microsoft	
Volume	count	2.516000e+03	2.516000e+03	2.516000e+03	2.516000e+03	2.5160
	mean	7.614520e+07	1.170867e+08	2.288008e+07	2.887935e+07	8.9042
	std	4.061943e+07	6.839957e+07	1.489149e+07	1.336292e+07	8.0183
	min	1.500750e+07	2.323470e+07	4.726100e+06	7.164500e+06	1.1440
	25%	5.071022e+07	7.105610e+07	1.440050e+07	2.060978e+07	4.2654
	50%	6.526200e+07	1.003646e+08	1.921050e+07	2.572470e+07	6.5676
	75%	9.008800e+07	1.426216e+08	2.639298e+07	3.315840e+07	1.0854
	max	4.771220e+08	6.488252e+08	2.323166e+08	1.691640e+08	1.3338
Close in \$	count	2.516000e+03	2.516000e+03	2.516000e+03	2.516000e+03	2.5160
	mean	1.012331e+02	9.636397e+01	2.222483e+02	1.877021e+02	3.3598
	std	5.417389e+01	6.525523e+01	1.216880e+02	1.228867e+02	1.8979
	min	1.434750e+01	2.258500e+01	7.405000e+01	4.029000e+01	4.5547
	25%	4.839713e+01	3.763938e+01	1.364025e+02	7.108000e+01	1.6224
	50%	9.457250e+01	6.666125e+01	1.830900e+02	1.534350e+02	3.3153
	75%	1.538562e+02	1.527725e+02	2.808725e+02	2.834975e+02	4.8144
	max	2.329300e+02	2.590200e+02	6.326800e+02	4.675600e+02	9.3656
Open in \$	count	2.516000e+03	2.516000e+03	2.516000e+03	2.516000e+03	2.5160
	mean	1.012573e+02	9.627565e+01	2.221506e+02	1.876574e+02	3.3588
	std	5.420085e+01	6.517499e+01	1.217244e+02	1.228868e+02	1.8974
	min	1.431400e+01	2.250000e+01	7.404000e+01	4.034000e+01	4.6021
	25%	4.858350e+01	3.760688e+01	1.360175e+02	7.088750e+01	1.6289
	50%	9.455800e+01	6.655000e+01	1.830600e+02	1.519800e+02	3.3136
	75%	1.538754e+02	1.525725e+02	2.791675e+02	2.838700e+02	4.8072

	Stock_Name	Amazon	Apple	Meta	Microsoft	
High in \$	max	2.323900e+02	2.581900e+02	6.315000e+02	4.670000e+02	9.3335
	count	2.516000e+03	2.516000e+03	2.516000e+03	2.516000e+03	2.5160
	mean	1.024060e+02	9.729970e+01	2.249749e+02	1.894414e+02	3.4079
	std	5.481932e+01	6.584931e+01	1.231828e+02	1.239561e+02	1.9190
	min	1.453950e+01	2.291750e+01	7.483000e+01	4.074000e+01	4.6785
	25%	4.882825e+01	3.804000e+01	1.378025e+02	7.160250e+01	1.6383
	50%	9.555600e+01	6.728125e+01	1.852000e+02	1.554800e+02	3.3631
	75%	1.559061e+02	1.545650e+02	2.852325e+02	2.870450e+02	4.8757
Low in \$	max	2.330000e+02	2.601000e+02	6.384000e+02	4.683500e+02	9.4175
	count	2.516000e+03	2.516000e+03	2.516000e+03	2.516000e+03	2.5160
	mean	1.000044e+02	9.532394e+01	2.193914e+02	1.857868e+02	3.3088
	std	5.352663e+01	6.456779e+01	1.201604e+02	1.217141e+02	1.8756
	min	1.426250e+01	2.236750e+01	7.200000e+01	3.972000e+01	4.5264
	25%	4.808713e+01	3.729563e+01	1.342275e+02	7.032250e+01	1.6041
	50%	9.345250e+01	6.580750e+01	1.811650e+02	1.509650e+02	3.2638
	75%	1.517300e+02	1.508125e+02	2.762875e+02	2.801550e+02	4.7585
MA_Six_Months	max	2.280100e+02	2.576300e+02	6.272100e+02	4.644600e+02	9.2308
	count	2.334000e+03	2.334000e+03	2.334000e+03	2.334000e+03	2.3340
	mean	1.007185e+02	9.376072e+01	2.147578e+02	1.839665e+02	3.2962
	std	4.892357e+01	6.013550e+01	9.786487e+01	1.136077e+02	1.5911
	min	2.119530e+01	2.526232e+01	8.384328e+01	4.463388e+01	8.3813
	25%	5.187800e+01	3.979064e+01	1.522523e+02	7.621671e+01	1.7869
	50%	9.546039e+01	6.682662e+01	1.816482e+02	1.556139e+02	3.3507
	75%	1.468468e+02	1.532224e+02	2.689032e+02	2.769961e+02	4.6211
MA_One_Year	max	1.905904e+02	2.173302e+02	5.302277e+02	4.253760e+02	7.0955
	count	2.152000e+03	2.152000e+03	2.152000e+03	2.152000e+03	2.1520
	mean	1.008385e+02	9.227470e+01	2.095900e+02	1.803159e+02	3.2823
	std	4.414964e+01	5.666534e+01	7.408294e+01	1.029689e+02	1.3651
	min	2.601075e+01	2.688353e+01	9.539885e+01	4.847038e+01	9.3695
	25%	6.094927e+01	4.112423e+01	1.697390e+02	8.243737e+01	2.2982
	50%	1.078959e+02	7.071487e+01	1.901022e+02	1.614129e+02	3.3639
	75%	1.396409e+02	1.538369e+02	2.574615e+02	2.770501e+02	4.3270
	max	1.701184e+02	1.997576e+02	4.490463e+02	3.964630e+02	5.9617

Step 4: Prepping for Visualizations

```
In [5]: # Grabbing today's date
max_date = max(stocks['Date'])
max_date
```

Out[5]: '2024-12-31'

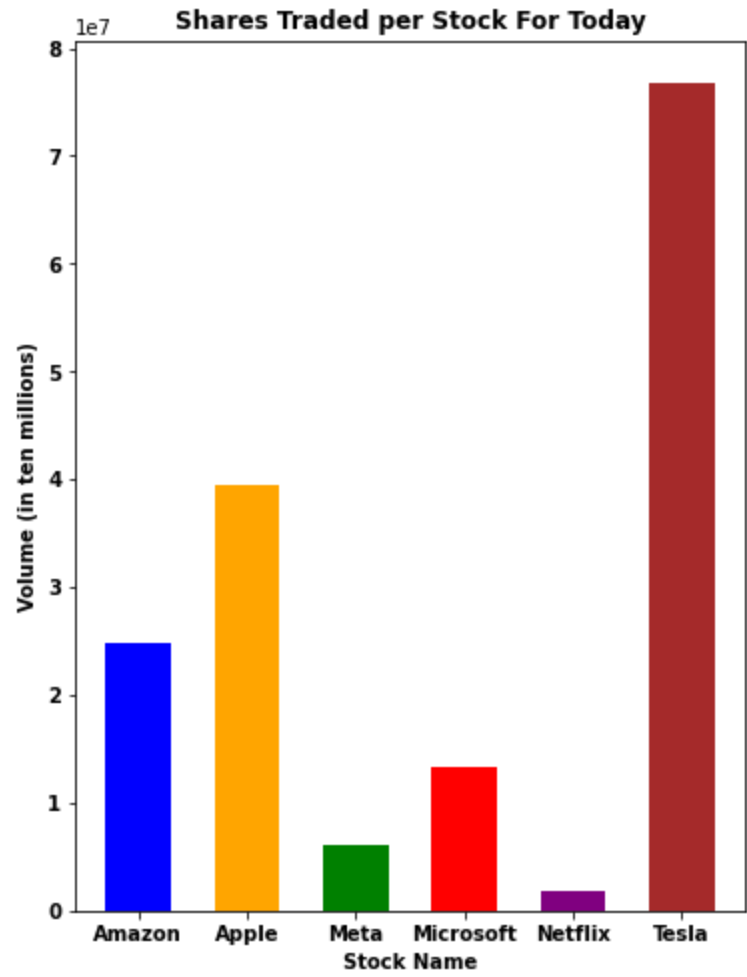
```
In [6]: # Filtering the dataset by using today's date
stocks_max_date = stocks[stocks['Date']==max_date]
stocks_max_date
```

Out[6]:

	Date	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One
2515	2024-12-31	Amazon	24819700	219.39	222.97	223.23	218.94	190.590383	170.1
5031	2024-12-31	Apple	39480700	250.42	252.44	253.28	249.43	217.330164	199.7
7547	2024-12-31	Meta	6019500	585.51	592.27	593.97	583.85	530.227650	449.04
10063	2024-12-31	Microsoft	13246500	421.50	426.10	426.73	420.66	425.358415	396.46
12579	2024-12-31	Netflix	1875900	891.32	901.80	902.68	889.47	709.552623	596.17
15095	2024-12-31	Tesla	76825100	403.84	423.79	427.93	402.54	244.968579	234.65

```
In [7]: shares_volume = plt.figure(figsize=(6,8))
colors = ['blue','orange','green','red','purple','brown']
plt.bar(stocks_max_date['Stock_Name'], stocks_max_date['Volume'],
        color = colors, width = 0.6),
plt.xlabel('Stock Name', fontsize = 10, fontweight = 'bold')
plt.ylabel('Volume (in ten millions)', fontsize = 10, fontweight = 'bold')
plt.xticks(fontsize = 10, rotation = 0, fontweight = 'bold')
plt.yticks(fontsize = 10, fontweight = 'bold')
plt.title('Shares Traded per Stock For Today', fontsize = 12, fontweight = 'bold')
```

Out[7]: Text(0.5, 1.0, 'Shares Traded per Stock For Today')



```
In [8]: stocks['Date'] = pd.to_datetime(stocks['Date'])
stocks = stocks.set_index('Date')
stocks
```

Out[8]:

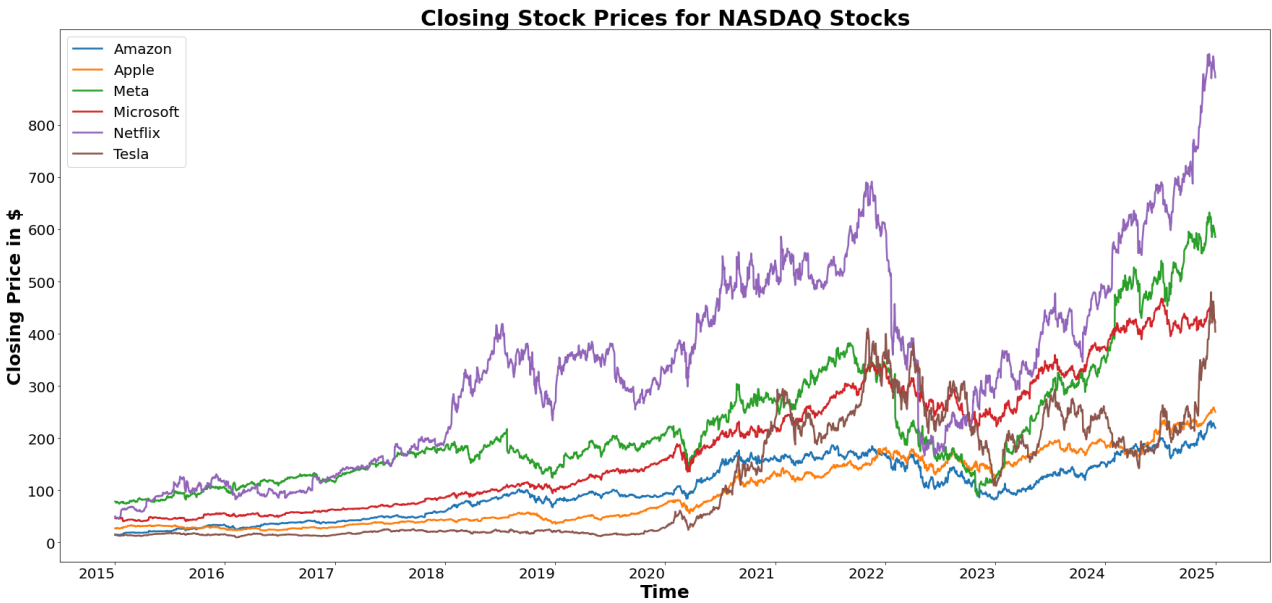
	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_O
Date								
2015-01-02	Amazon	55664000	15.4260	15.6290	15.7375	15.3480	NaN	
2015-01-05	Amazon	55484000	15.1095	15.3505	15.4190	15.0425	NaN	
2015-01-06	Amazon	70380000	14.7645	15.1120	15.1500	14.6190	NaN	
2015-01-07	Amazon	52806000	14.9210	14.8750	15.0640	14.7665	NaN	
2015-01-08	Amazon	61768000	15.0230	15.0160	15.1570	14.8055	NaN	
...	...	...	...	...	...	...	...	
2024-12-24	Tesla	59551800	462.2800	435.9000	462.7800	435.1400	239.391967	233

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_O
Date								
2024-12-26	Tesla	76366400	454.1300	465.1600	465.3300	451.0200	240.972459	233
2024-12-27	Tesla	82666800	431.6600	449.5200	450.0000	426.5000	242.386011	233
2024-12-30	Tesla	64941000	417.4100	419.4000	427.0000	415.7500	243.700383	234
2024-12-31	Tesla	76825100	403.8400	423.7900	427.9300	402.5400	244.968579	234.

15096 rows × 8 columns

```
In [9]: #Using a line chart to graph closing price of stocks within a 10 year frame
six_stocks_ten_years = plt.figure(figsize=(30,15))
stocks.groupby('Stock_Name')['Close in $'].plot(legend=True, linewidth=2.5)
plt.title('Closing Stock Prices for NASDAQ Stocks', fontsize = 30, fontweight =
plt.xlabel('Time', fontsize = 25, fontweight = 'bold')
plt.xticks(rotation = 0, fontsize = 20)
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 900, 100)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

Out[9]: <matplotlib.legend.Legend at 0x7f8c19871670>



```
In [10]: # Breaking dataset into multiple where each one is assigned to one specific stock
```

```
In [11]: moving_average = stocks[stocks['MA_One_Year'].notna()]
moving_average
```

Out[11]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_O
Date								
2016-06-14	Amazon	50138000	35.9650	35.6165	36.0405	35.6135	30.830541	26
2016-06-15	Amazon	54188000	35.7130	36.1000	36.1280	35.6675	30.879227	26
2016-06-16	Amazon	61960000	35.8755	35.6025	35.9000	35.2650	30.929434	26
2016-06-17	Amazon	117956000	35.3195	35.9095	35.9100	34.9590	30.979199	26
2016-06-20	Amazon	73544000	35.7005	35.6750	36.0655	35.5405	31.036563	26
...	...	...	...	...	...	...	...	...
2024-12-24	Tesla	59551800	462.2800	435.9000	462.7800	435.1400	239.391967	233
2024-12-26	Tesla	76366400	454.1300	465.1600	465.3300	451.0200	240.972459	233
2024-12-27	Tesla	82666800	431.6600	449.5200	450.0000	426.5000	242.386011	233
2024-12-30	Tesla	64941000	417.4100	419.4000	427.0000	415.7500	243.700383	234
2024-12-31	Tesla	76825100	403.8400	423.7900	427.9300	402.5400	244.968579	234

12912 rows x 8 columns

In [12]:

```
moving_average_amazon_stock = moving_average[moving_average['Stock_Name']=='Amazon']
moving_average_amazon_stock
```

Out[12]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_O
Date								
2016-06-14	Amazon	50138000	35.9650	35.6165	36.0405	35.6135	30.830541	26
2016-06-15	Amazon	54188000	35.7130	36.1000	36.1280	35.6675	30.879227	26.
2016-06-16	Amazon	61960000	35.8755	35.6025	35.9000	35.2650	30.929434	26
2016-06-17	Amazon	117956000	35.3195	35.9095	35.9100	34.9590	30.979199	26
2016-06-20	Amazon	73544000	35.7005	35.6750	36.0655	35.5405	31.036563	26
...	...	...	...	...	...	...	...	...

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_O
Date								
2024-12-24	Amazon	15007500	229.0500	226.9400	229.1400	226.1300	189.772842	169
2024-12-26	Amazon	16146700	227.0500	228.5000	228.5000	226.6700	190.002240	169.
2024-12-27	Amazon	27367100	223.7500	225.6000	226.0300	220.9000	190.212951	169.
2024-12-30	Amazon	28321200	221.3000	220.0600	223.0000	218.4300	190.407650	169
2024-12-31	Amazon	24819700	219.3900	222.9700	223.2300	218.9400	190.590383	170

2152 rows × 8 columns

In [13]:

```
moving_average_apple_stock = moving_average[moving_average['Stock_Name']=='Apple']
moving_average_apple_stock
```

Out[13]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_C
Date								
2016-06-14	Apple	127727600	24.3650	24.3300	24.6200	24.1875	26.352350	28
2016-06-15	Apple	117780800	24.2850	24.4550	24.6025	24.2575	26.328880	28
2016-06-16	Apple	125307200	24.3875	24.1125	24.4375	24.0175	26.305041	28
2016-06-17	Apple	244032800	23.8325	24.1550	24.1625	23.8250	26.278566	28
2016-06-20	Apple	137647600	23.7750	24.0000	24.1425	23.7575	26.254877	28
...	...	...	...	...	...	...	...	...
2024-12-24	Apple	23234700	258.2000	255.4900	258.2100	255.2900	215.462678	199
2024-12-26	Apple	27237100	259.0200	258.1900	260.1000	257.6300	215.951421	199
2024-12-27	Apple	42355300	255.5900	257.8300	258.7000	253.0600	216.427596	199
2024-12-30	Apple	35557500	252.2000	252.2300	253.5000	250.7500	216.878579	199
2024-12-31	Apple	39480700	250.4200	252.4400	253.2800	249.4300	217.330164	199

2152 rows × 8 columns

In [14]:

```
moving_average_meta_stock = moving_average[moving_average['Stock_Name']=='Meta']
moving_average_meta_stock
```

Out[14]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
Date								
2016-06-14	Meta	17618500	114.94	114.07	114.95	113.58	106.946612	95.398849
2016-06-15	Meta	19819400	114.60	115.30	115.44	114.07	107.059344	95.497890
2016-06-16	Meta	19236300	114.39	113.87	114.50	112.94	107.168525	95.599808
2016-06-17	Meta	24644300	113.02	114.42	114.43	112.56	107.279180	95.700822
2016-06-20	Meta	20785400	113.37	113.77	114.72	112.75	107.411202	95.802795
...	...	...	...	...	...	...	...	...
2024-12-24	Meta	4726100	607.75	602.72	607.99	599.28	528.606940	445.926575
2024-12-26	Meta	6081400	603.35	605.48	606.30	598.94	529.022295	446.728575
2024-12-27	Meta	8084200	599.81	599.41	601.85	589.80	529.462514	447.516959
2024-12-30	Meta	7025900	591.24	588.75	596.94	585.58	529.868743	448.271014
2024-12-31	Meta	6019500	585.51	592.27	593.97	583.85	530.227650	449.046329

2152 rows x 8 columns

In [15]:

```
moving_average_microsoft_stock = moving_average[moving_average['Stock_Name']=='Microsoft']
moving_average_microsoft_stock
```

Out[15]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
Date								
2016-06-14	Microsoft	42577100	49.83	49.90	50.10	49.57	52.187049	48.470384
2016-06-15	Microsoft	33757600	49.69	49.78	50.12	49.69	52.218852	48.478411
2016-06-16	Microsoft	31188600	50.39	49.52	50.47	49.51	52.254262	48.489534
2016-06-17	Microsoft	45710500	50.13	50.41	50.43	49.82	52.288087	48.501808

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
Date								
2016-06-20	Microsoft	35607900	50.07	50.64	50.83	50.03	52.325137	48.512329
...	...	...	...	...	...	...	...	...
2024-12-24	Microsoft	7164500	439.33	434.65	439.60	434.19	425.274699	395.619726
2024-12-26	Microsoft	8194200	438.11	439.08	440.94	436.63	425.343497	395.872822
2024-12-27	Microsoft	18117700	430.53	434.60	435.22	426.35	425.375956	396.067452
2024-12-30	Microsoft	13158700	424.83	426.06	427.55	421.90	425.368033	396.258548
2024-12-31	Microsoft	13246500	421.50	426.10	426.73	420.66	425.358415	396.463014

2152 rows × 8 columns

In [16]:

```
moving_average_netflix_stock = moving_average[moving_average['Stock_Name']=='Net  
moving_average_netflix_stock
```

Out[16]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
Date								
2016-06-14	Netflix	9480400	94.12	94.16	95.80	93.27	103.602240	93.695698
2016-06-15	Netflix	7675400	94.29	94.61	95.46	93.50	103.581585	93.817455
2016-06-16	Netflix	7550800	95.44	94.18	95.56	93.25	103.536120	93.949314
2016-06-17	Netflix	8046800	94.45	95.75	95.77	94.20	103.493552	94.080680
2016-06-20	Netflix	9026400	93.80	95.22	95.88	93.65	103.462568	94.209604
...	...	...	...	...	...	...	...	...
2024-12-24	Netflix	2320300	932.12	915.00	935.85	911.70	703.421093	591.288411
2024-12-26	Netflix	2340300	924.14	928.40	930.49	915.30	704.994645	592.587288
2024-12-27	Netflix	3226200	907.55	916.01	918.13	894.50	706.520000	593.772904
2024-12-30	Netflix	2203000	900.43	894.51	908.23	889.71	708.062240	594.931370

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
Date								
2024-12-31	Netflix	1875900	891.32	901.80	902.68	889.47	709.552623	596.174932

2152 rows × 8 columns

In [17]:

```
moving_average_tesla_stock = moving_average[moving_average['Stock_Name']=='Tesla']
moving_average_tesla_stock
```

Out[17]:

	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_O
Date								
2016-06-14	Tesla	53703000	14.3307	14.5920	14.8133	14.1687	14.561533	14
2016-06-15	Tesla	43627500	14.5133	14.4633	14.7933	14.3420	14.545737	14.
2016-06-16	Tesla	36604500	14.5287	14.4947	14.5360	14.2333	14.529275	14
2016-06-17	Tesla	46689000	14.3647	14.5207	14.6660	14.3000	14.514179	14
2016-06-20	Tesla	53332500	14.6467	14.6333	14.9167	14.5487	14.503713	14
...	...	...	...	...	...	...	...	...
2024-12-24	Tesla	59551800	462.2800	435.9000	462.7800	435.1400	239.391967	233
2024-12-26	Tesla	76366400	454.1300	465.1600	465.3300	451.0200	240.972459	233
2024-12-27	Tesla	82666800	431.6600	449.5200	450.0000	426.5000	242.386011	233
2024-12-30	Tesla	64941000	417.4100	419.4000	427.0000	415.7500	243.700383	234
2024-12-31	Tesla	76825100	403.8400	423.7900	427.9300	402.5400	244.968579	234.

2152 rows × 8 columns

In [18]:

```
#Using a line chart to graph one year and five year moving averages for all stock
amazon = plt.figure(figsize=(30,15))
moving_average_amazon_stock.groupby('Stock_Name')['Close in $'].plot(legend=True)
moving_average_amazon_stock.groupby('Stock_Name')['MA_Six_Months'].plot(legend=True)
moving_average_amazon_stock.groupby('Stock_Name')['MA_One_Year'].plot(legend=True)
plt.title('Closing Price Per Share For Amazon', fontsize = 30, fontweight = 'bold')
plt.xlabel('Time', fontsize = 25, fontweight = 'bold')
plt.xticks(rotation = 0, fontsize = 20)
```

```
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 300, 50)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

Out[18]: <matplotlib.legend.Legend at 0x7f8be084ed00>



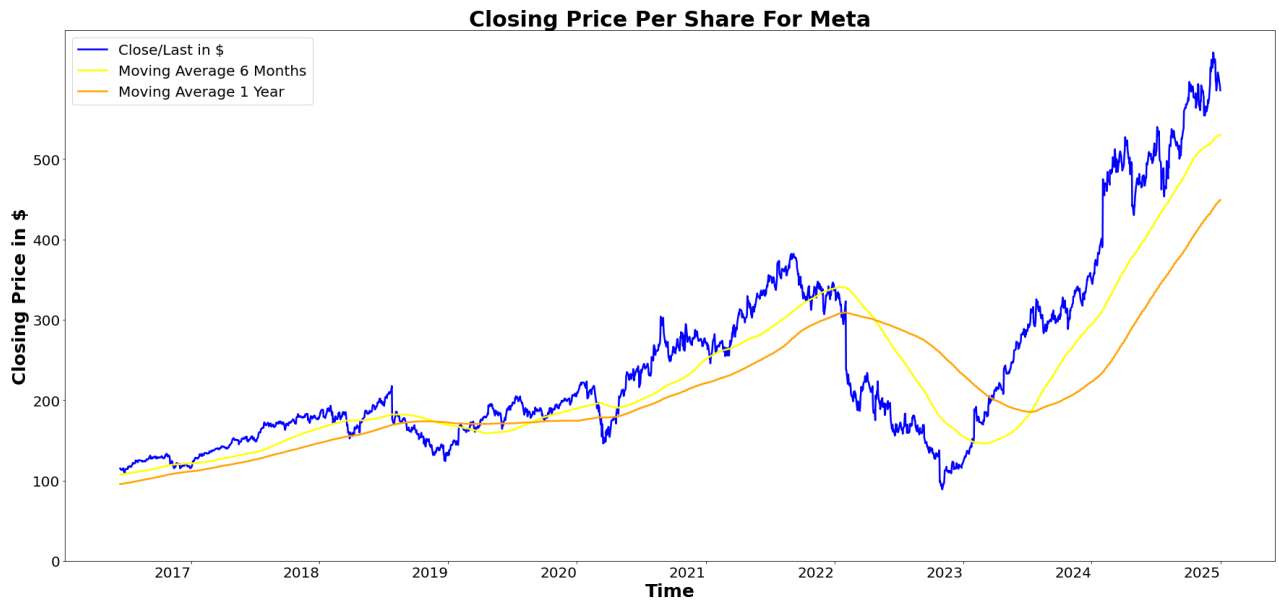
```
In [19]: apple = plt.figure(figsize=(30,15))
moving_average_apple_stock.groupby('Stock_Name')['Close in $'].plot(legend=True,
moving_average_apple_stock.groupby('Stock_Name')['MA_Six_Months'].plot(legend=Tr
moving_average_apple_stock.groupby('Stock_Name')['MA_One_Year'].plot(legend=True
plt.title('Closing Price Per Share For Apple', fontsize = 30, fontweight = 'bold')
plt.xlabel('Time', fontsize = 25, fontweight = 'bold')
plt.xticks(rotation = 0, fontsize = 20)
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 300, 50)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

Out[19]: <matplotlib.legend.Legend at 0x7f8c38b65ac0>



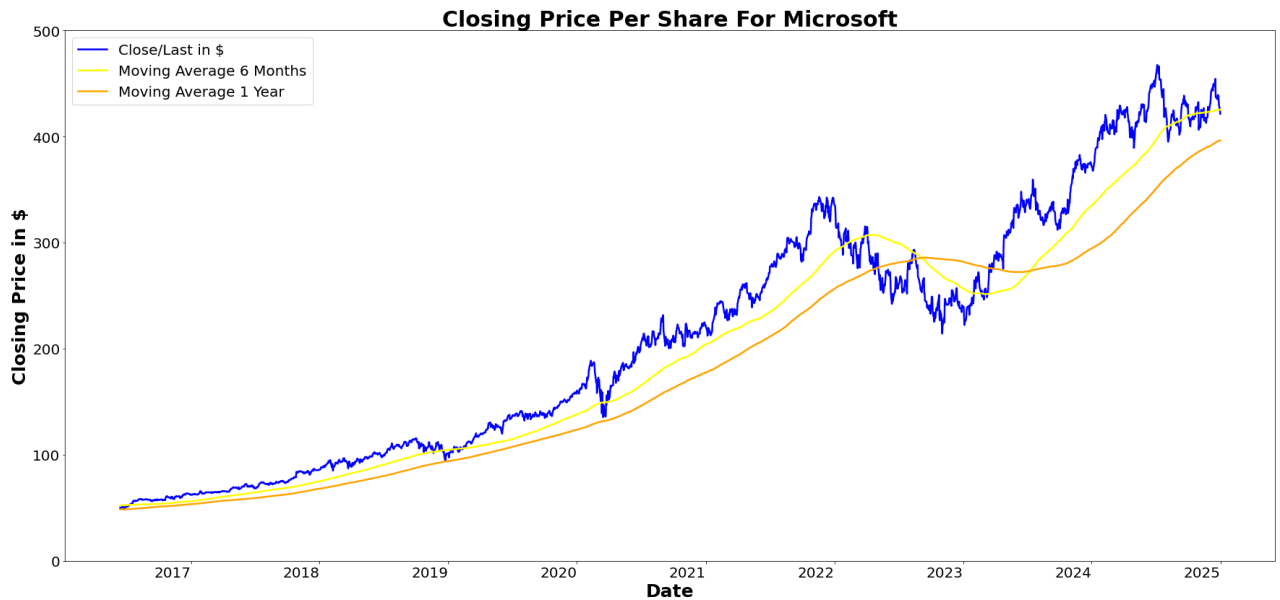
```
In [20]: meta = plt.figure(figsize=(30,15))
moving_average_meta_stock.groupby('Stock_Name')['Close in $'].plot(legend=True,
moving_average_meta_stock.groupby('Stock_Name')['MA_Six_Months'].plot(legend=Tru
moving_average_meta_stock.groupby('Stock_Name')['MA_One_Year'].plot(legend=True,
plt.title('Closing Price Per Share For Meta', fontsize = 30, fontweight = 'bold')
plt.xlabel('Time', fontsize = 25, fontweight = 'bold')
plt.xticks(rotation = 0, fontsize = 20)
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 600, 100)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

Out[20]: <matplotlib.legend.Legend at 0x7f8c38cac880>



```
In [21]: microsoft = plt.figure(figsize=(30,15))
moving_average_microsoft_stock.groupby('Stock_Name')['Close in $'].plot(legend=T
moving_average_microsoft_stock.groupby('Stock_Name')['MA_Six_Months'].plot(leger
moving_average_microsoft_stock.groupby('Stock_Name')['MA_One_Year'].plot(legend=
plt.title('Closing Price Per Share For Microsoft', fontsize = 30, fontweight = '
plt.xlabel('Date', fontsize = 25, fontweight = 'bold')
plt.xticks(rotation = 0, fontsize = 20)
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 600, 100)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

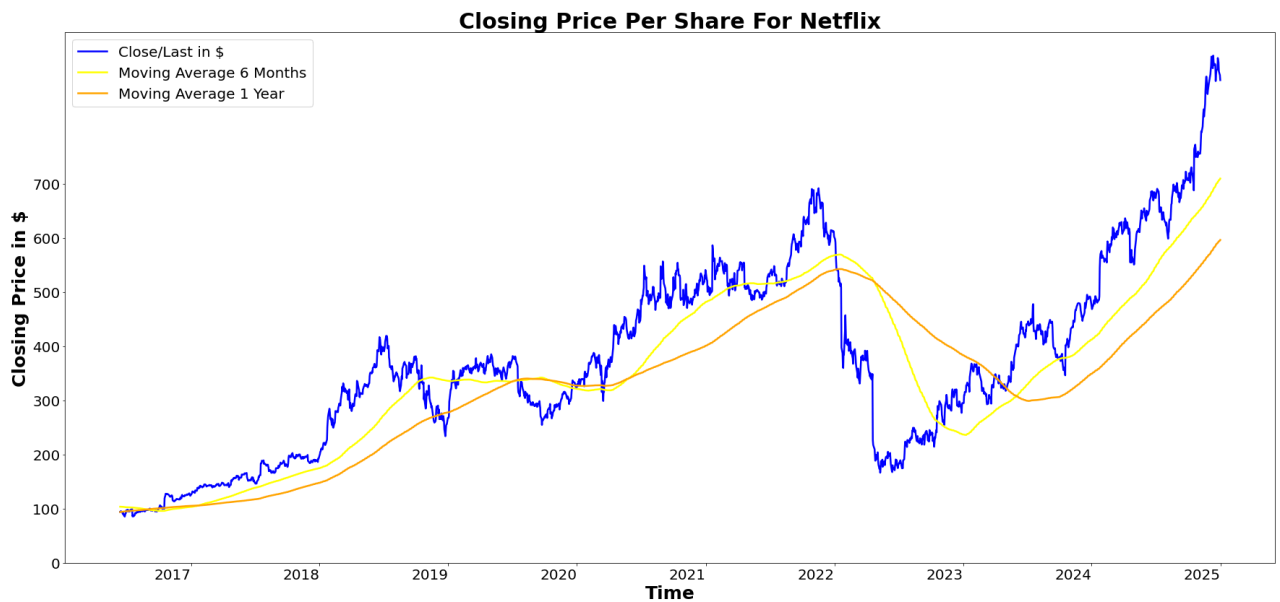
Out[21]: <matplotlib.legend.Legend at 0x7f8c0854fb50>



In [22]:

```
netflix = plt.figure(figsize=(30,15))
moving_average_netflix_stock.groupby('Stock_Name')['Close in $'].plot(legend=True)
moving_average_netflix_stock.groupby('Stock_Name')['MA_Six_Months'].plot(legend=True)
moving_average_netflix_stock.groupby('Stock_Name')['MA_One_Year'].plot(legend=True)
plt.title('Closing Price Per Share For Netflix', fontsize = 30, fontweight = 'bold')
plt.xlabel('Time', fontsize = 25, fontweight = 'bold')
plt.xticks(rotation = 0, fontsize = 20)
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 800, 100)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

Out[22]: <matplotlib.legend.Legend at 0x7f8c0853d400>



In [23]:

```
tesla = plt.figure(figsize=(30,15))
moving_average_tesla_stock.groupby('Stock_Name')['Close in $'].plot(legend=True,
moving_average_tesla_stock.groupby('Stock_Name')['MA_Six_Months'].plot(legend=True)
moving_average_tesla_stock.groupby('Stock_Name')['MA_One_Year'].plot(legend=True)
plt.title('Closing Price Per Share For Tesla', fontsize = 30, fontweight = 'bold')
plt.xlabel('Time', fontsize = 25, fontweight = 'bold')
```

```
plt.xticks(rotation = 0, fontsize = 20)
plt.ylabel('Closing Price in $', fontsize = 25, fontweight = 'bold')
y_ticks = np.arange(0, 500, 100)
plt.yticks(y_ticks, rotation = 0, fontsize = 20)
plt.legend(fontsize=20)
```

Out[23]: <matplotlib.legend.Legend at 0x7f8c1a5625b0>



```
In [24]: shares_volume.savefig(os.path.join(path, 'Visualizations', 'shares_volume.png'))
six_stocks_ten_years.savefig(os.path.join(path, 'Visualizations', 'six_stocks_te
amazon.savefig(os.path.join(path, 'Visualizations', 'amazon.png'))
apple.savefig(os.path.join(path, 'Visualizations', 'apple.png'))
meta.savefig(os.path.join(path, 'Visualizations', 'meta.png'))
microsoft.savefig(os.path.join(path, 'Visualizations', 'microsoft.png'))
netflix.savefig(os.path.join(path, 'Visualizations', 'netflix.png'))
tesla.savefig(os.path.join(path, 'Visualizations', 'tesla.png'))
```